SABER: Support-Conditioned Bellman Error Allocation after Hindsight Relabeling

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Abstract

Hindsight Experience Replay (HER) converts failed trajectories into useful goal-conditioned Bellman targets, but it does not determine how a fixed critic update budget should be allocated among the resulting residuals. We study this post-relabeling allocation problem and introduce **SABER**, a critic-only objective that assigns more fitting pressure when the current policy action is represented in a task-local replay-action support set. SABER leaves hindsight goal selection, replay sampling, Bellman targets, actor objectives, and actor-critic architectures unchanged. Under matched one-update-per-step training, SABER raises mean success AUC over HER on FetchPickAndPlace with DDPG, TD3, and SAC. A paired 10-seed TD3 confirmation yields an AUC difference of 0.0467 with a 95% paired bootstrap interval of [0.0085, 0.0783]. On sparse DrawerOpen with DDPG, mean AUC rises from 0.207 to 0.754 and Hit90 coverage from 2/5 to 5/5 seeds. Controls show a narrower mechanism boundary: SABER is not equivalent to TD-error goal ranking, but count-preserving critic reweighting and larger update budgets can be competitive, and a context-eligible count-matched control does not establish an independent membership contribution at five seeds. Diagnostics show that support-positive samples are enriched among high-error updates and receive larger immediate residual correction, but are not lower-variance targets. SABER therefore exposes a useful critic-loss allocation axis after hindsight relabeling; it is not a universal improvement or a wall-clock efficiency claim.

1. Introduction

Goal-conditioned reinforcement learning learns a policy that must solve many goals rather than maximize a single fixed reward. In sparse robotic manipulation, successful transitions are rare, so a replay buffer collected under the original goal contains little direct positive supervision. Hindsight Experience Replay (HER) addresses this bottleneck by relabeling failed trajectories with goals that were achieved later in the same episode (Andrychowicz et al. 2017). This converts otherwise unsuccessful experience into valid goal-conditioned Bellman targets and has become a standard component of off-policy goal-conditioned learning.

Relabeling, however, answers only one of two decisions. It determines **which targets are constructed**, but not **how strongly the critic should fit each sample once a minibatch has been formed**. Standard HER actor-critics generally assign equal critic-loss weight to all sampled relabeled transitions. Existing HER variants instead change the goal distribution, episode distribution, or target construction; prioritized replay changes sampling frequency according to surprise. We ask an orthogonal post-relabeling question:

> *Conditional on the same sampled HER minibatch and the same Bellman targets, should every critic residual receive the same fitting pressure?*

This distinction does not imply that HER targets are invalid. It separates target construction from loss allocation. In an off-policy actor-critic, the critic is regressed at replay actions while the actor queries the critic near the current policy action. When that policy action is represented by replay actions from a similar state-goal-horizon context, fitting the corresponding Bellman residual more strongly may better allocate a fixed critic update. When it is not represented, the same residual remains valid training data, but its connection to the action region currently exploited by the actor is less direct. This is a design hypothesis, not a theorem that supported targets are intrinsically cleaner.

We instantiate the hypothesis with **Support-Conditioned Adaptive Bellman Error Reweighting (SABER)**. For each sampled transition, SABER builds a discrete local context, retrieves replay transitions in that context, and checks whether the current policy action bin appears among their replay-action bins. A binary support predicate raises the sample's raw critic weight, after which weights are normalized to unit batch mean. The main method is deliberately critic-only: HER relabeling, uniform replay, Bellman targets, actor losses, exploration, and backbone networks remain unchanged. This makes SABER a modular loss objective, not a new replay algorithm or actor-critic backbone.

Our evaluation is organized around three hypotheses rather than a benchmark leaderboard. First, we test whether SABER can improve matched HER under DDPG, TD3, and SAC. Second, we test alternative explanations using Hindsight Goal Ranking (HGR), TD-error and prioritized replay baselines, random and count-matched assignments, high replay ratios, and REDQ-HER. Third, we inspect the selected samples through TD-error enrichment, immediate residual correction, target variance, phase allocation, failure cases, and runtime. This organization also makes negative results part of the claim: SABER loses on FetchPush SAC and FetchSlide DDPG, multiple random controls are competitive in some settings, and support lookup has substantial implementation cost.

Our contributions are:

1. We isolate **post-relabeling critic-loss allocation** as a distinct design problem in sparse HER actor-critic learning.
2. We introduce SABER, a critic-only support-conditioned objective that changes loss allocation while holding goal relabeling, replay, targets, actors, and backbone architectures fixed.
3. We provide controlled evidence across tasks and backbones, including a paired 10-seed confirmation, a matched HGR baseline, compute-scaled references, count-matched controls, mechanism diagnostics, and explicit failure and runtime analyses.

2. Related Work

Hindsight Relabeling and Goal Prioritization

HER relabels failed trajectories using achieved goals (Andrychowicz et al. 2017). Later methods improve the selection or construction of hindsight goals. Hindsight Goal Generation chooses goals that connect achieved states to desired goals (Ren et al. 2019); Curriculum-Guided HER balances proximity and diversity over training (Fang et al. 2019); energy-based prioritization ranks trajectories by movement energy (Zhao and Tresp 2018); HGR ranks episodes and future-goal pairs using TD error (Luu and Yoo 2021); and multi-step HER changes the return used for relabeled targets (Yang et al. 2021). These methods intervene before or during target selection. SABER keeps the future-goal HER distribution fixed and operates after relabeling, when the critic loss is assembled. The two directions are therefore complementary.

Replay and Bellman-Loss Reweighting

Prioritized Experience Replay (PER) changes replay frequency according to TD error (Schaul et al. 2016). Attentive Experience Replay instead prioritizes states related to the current policy's visitation pattern (Sun, Zhou, and Li 2020). Covariate-shift corrections reweight off-policy updates toward a target stationary distribution (Gelada and Bellemare 2019). DisCor derives a training distribution intended to correct propagated value error and downweights targets estimated to be inaccurate (Kumar, Gupta, and Levine 2020). SABER belongs to this broad critic-allocation family but differs in signal and scope: it does not learn an error model, estimate a density ratio, or alter replay sampling. It uses task-local current-policy action membership as a coarse binary signal inside sparse HER minibatches.

Goal-Conditioned Critic Objectives and Compute Scaling

ReenGAGE augments goal-conditioned Q-learning with a knowledge-distillation objective that matches goal gradients of current and target critics (Levine and Feizi 2023). Like SABER, it demonstrates that the critic objective can be modified while retaining a DDPG+HER backbone, but it transfers information across goal dimensions rather than allocating residual weight by action support. DDPG, TD3, and SAC provide the backbones used here (Lillicrap et al. 2015; Fujimoto, van Hoof, and Meger 2018; Haarnoja et al. 2018). High replay ratios and REDQ increase the number of updates or critics (Chen et al. 2021; Hiraoka 2023). We include these as compute- and architecture-scaled references, not as matched-cost instances of SABER.

3. Post-Relabeling Critic-Loss Allocation

Consider a goal-conditioned MDP $\mathcal{M}=(\mathcal{S},\mathcal{A},\mathcal{G},P,r_g,\gamma)$. A replay transition contains observation o_i, replay action a_i^r, next observation o'_i, original goal, timeout information, and horizon context. HER samples a training goal g_i from the episode's future achieved goals with probability 0.8 in our experiments. Let $x_i=(o_i,g_i)$ and $x'_i=(o'_i,g_i)$. The backbone constructs its ordinary target y_i; for a deterministic single-critic backbone,

$$y_i = r(o'_i, g_i) + \gamma(1 - d_i)\, Q_{\varphi\text{bar}}(x'_i, \pi_{\theta\text{bar}}(x'_i))$$

with backbone-specific twin-critic, target-smoothing, or entropy terms for TD3 and SAC. The standard critic objective is

$$L_Q^{HER}(\varphi) = (1/B) \Sigma_{i=1}^{B} [Q_\varphi(x_i, a_i^r) - y_i]^2$$

HER determines g_i and hence y_i, while this objective allocates equal weight after the minibatch has been sampled. Our problem is to construct a sample weight using information that is distinct from reward relabeling and then test whether it improves learning under a fixed update budget.

4. SABER

Local Context and Replay-Action Support

For sample i, a task adapter maps the observation, relabeled training goal, and horizon to a discrete context

$$c_i = C(o_i, g_i, h_i).$$

The context encodes a coarse manipulation phase and binned task variables, such as hand-object distance, object-goal distance, height, gripper state, and horizon. The local pool is

$$L_i = \{j \in D : C(o_j, g_i, h_j) = C(o_i, g_i, h_i)\}.$$

Replay transition j is re-contextualized under the query sample's training goal g_i, rather than under its original environment goal. For efficiency, the implementation retrieves a deterministic tail of at most N_{scan} candidates from a coarse context bucket and applies the full context predicate to that subset.

Each continuous action dimension is partitioned into $K=5$ bins, and the per-dimension bins are combined into a mixed-radix action-bin identifier. Let

$$B_i = \{b(a_j^r) : j \in L_i\}, \; a_i^\pi = \pi_\theta(o_i, g_i).$$

DDPG and TD3 use their deterministic actor output for a_i^{π}. SAC uses the deterministic tanh-mean action for the support query; its stochastic actor sampling and entropy term in the Bellman target remain unchanged.

The main support predicate is

$$\psi_i = 1[\ |L_i| \geq N_{local}\ \wedge\ \min(|L_i|, M_B) \geq M_{avail}\ \wedge\ b(a_i^{\pi}) \in B_i\ ].$$

We use $N_{\mathrm{local}}=4$, $M_B=16$, and $M_{\mathrm{avail}}=4$. Under these settings the count requirement largely reduces to having at least four local transitions; the discriminating term is current-policy action-bin membership. This is a coarse local-membership proxy, not a calibrated support density, a target-noise estimator, or an offline-RL constraint.

Critic-Only Weighting

SABER assigns raw weight

$$w_i^{raw} = 1 + \lambda_S \psi_i,$$

with $\lambda_S=1$, and normalizes the batch to unit mean:

$$\Omega_i = w_i^{raw} / [B^{-1} \Sigma_{j=1}^{B} w_j^{raw} + \varepsilon].$$

The critic loss becomes

$$L_Q^{SABER}(\varphi) = (1/B) \Sigma_{i=1}^{B} \Omega_i\ [Q_\varphi(x_i, a_i^{r}) - y_i]^2.$$

The same Ω_i weights both critics under TD3 and SAC. The actor loss is unchanged. Thus SABER preserves the minibatch, relabeled goals, rewards, terminal convention, Bellman targets, optimizer count, and actor objective.

What the Predicate Does and Does Not Mean

The critic residual is evaluated at a_i^{r}, whereas the predicate queries support for a_i^{π}. We therefore do not claim that ψ_i directly measures the noise of y_i at the replay action. The intended interpretation is narrower: ψ_i identifies contexts in which replay contains an action family that includes the action region currently queried by the actor. The experiments test whether allocating more critic pressure to this subset is useful, whether random/count-matched allocation explains the result, and how the subset relates to TD error and immediate correction.

Figure 1 (planned). A single intervention diagram should show: trajectory collection and HER target construction on the left; the unchanged sampled minibatch and Bellman targets in the center; SABER's context, policy-action query, replay-bin membership, and normalized weight above the critic residual; and an unchanged actor objective. Goal-selection methods, PER, and high replay ratio should be shown as different intervention points.

5. Experimental Design

Hypotheses

H1 (matched effect). Does SABER improve success AUC under a fixed HER backbone and one update per environment step?

H2 (attribution). Can the effect be explained by goal prioritization, TD-error weighting, random or count-preserving critic reweighting, or additional updates and critics?

H3 (mechanism and scope). Which samples receive weight, how are they related to TD error and residual correction, where does SABER fail, and what does the support lookup cost?

Tasks, Backbones, and Protocol

The primary breadth task is sparse FetchPickAndPlace-v4 at one million environment steps under DDPG, TD3, and SAC. Sparse Meta-World DrawerOpen-v3 at two million steps provides a harder DDPG large-effect case. Secondary scope tests use FetchPush, FetchSlide, DrawerOpen TD3, and DrawerClose DDPG/TD3.

Unless stated otherwise, runs use five seeds, batch size 256, replay capacity 300k, $\gamma=0.98$, $\tau=0.005$, actor and critic learning rates 3×10^{-4}, future HER ratio 0.8, 20 evaluation episodes, and one gradient update per environment step. Success states do not force termination in the target and time-limit transitions bootstrap. Actors and critics use two-layer 256-unit MLPs. The TD3 confirmation uses paired seeds 1--10 and an exact 10k-to-1M evaluation grid.

The primary metric is normalized trapezoidal success AUC. Final success and Hit90 coverage are secondary because sparse manipulation often saturates or collapses late. We report arithmetic means and sample standard deviations. Method comparisons use paired seed differences and percentile bootstrap 95% intervals. For the 10-seed TD3 confirmation we additionally report the exact two-sided sign-flip value. No interpolation, smoothing, forward filling, or seed replacement is used in the confirmatory and packaged control analyses.

6. Results

H1: SABER Improves Matched HER in the Primary Settings

Table 1 separates the original five-seed breadth comparison from the paired 10-seed TD3 confirmation. SABER raises mean AUC in all three FPP backbones. The five-seed DDPG estimate is uncertain, whereas the SAC interval and the 10-seed TD3 bootstrap interval exclude zero. For TD3, 8/10 paired seeds have a positive AUC difference; the exact two-sided sign-flip value is 0.0527, which should be read alongside, not replaced by, the bootstrap interval.

Table 1. Primary matched HER comparisons. AUC is normalized success AUC. Hit90 reports reached-seed counts. Deltas are SABER minus HER paired AUC.

Task / backbone	Pairs	HER AUC	SABER AUC	Paired AUC delta [95% CI]	Final, HER / SABER	Hit90, HER / SABER
FPP / DDPG	5	0.539 +/- 0.125	0.609 +/- 0.038	+0.070 [-0.016, 0.193]	0.940 / 0.960	5/5 / 5/5
FPP / TD3, confirmatory	10	0.383 +/- 0.041	0.429 +/- 0.058	+0.0467 [0.0085, 0.0783]	0.860 / 0.895	7/10 / 8/10
FPP / SAC	5	0.463 +/- 0.077	0.565 +/- 0.030	+0.102 [0.057, 0.150]	0.790 / 0.990	3/5 / 5/5
DrawerOpen / DDPG	5	0.207 +/- 0.287	0.754 +/- 0.044	+0.546 [0.340, 0.741]	0.390 / 1.000	2/5 / 5/5

DrawerOpen provides the largest practical effect. All five SABER seeds reach final success 1.0 and Hit90 between 400k and 580k, while HER succeeds in only two of five seeds. A separate diagnostic SABER cohort over seeds 1--10 reaches Hit90 in 9/10 seeds and has mean AUC 0.653. Because that cohort combines a seeds-1--5 diagnostic rerun with a seeds-6--10 extension and lacks a matched 10-seed HER rerun, it is robustness evidence rather than a new paired performance comparison.

Figure 2 (planned). The main learning-curve figure should contain four panels: FPP DDPG, FPP TD3, FPP SAC, and DrawerOpen DDPG. Thin seed curves should remain visible under a mean line and standard-deviation band. The TD3 panel should use all ten confirmatory pairs; the other panels should use the exact cohorts in Table 1.

H2: Controls Bound the Explanation

HGR is the closest goal-prioritization baseline. Our packaged FPP implementation uses completed-episode pair priorities, episode-then-pair sampling, and importance correction under the matched DDPG+HER backbone. HGR obtains AUC 0.338 +/- 0.149, below both HER (0.539 +/- 0.125) and SABER (0.609 +/- 0.038), with Hit90 in 2/5 seeds. This result shows that the SABER gain is not reproduced by this TD-error hindsight-goal prioritizer in FPP; it does not establish that critic allocation is universally superior to HGR.

The strongest assignment control is the FPP TD3 context-eligible count-matched experiment. It recomputes the same context-eligible set and the counterfactual SABER positive count at every update, then assigns exactly that many positive weights uniformly within the eligible set. Positive counts and raw and normalized weight histograms are therefore matched. SABER reaches AUC 0.442, compared with 0.391 for the control, but the paired difference 0.0516 has interval [-0.0165, 0.0992] and exact sign-flip value 0.25. Exact action-membership assignment is directionally helpful here but its independent contribution is not confirmed at five seeds.

Two query ablations provide complementary evidence. Replacing the current policy query with the sampled replay action gives AUC 0.368, and shuffling policy actions across samples gives 0.347, compared with 0.442 for SABER. Their paired bootstrap intervals against SABER exclude zero, although the smallest possible nonzero two-sided sign-flip value with five same-direction pairs is 0.0625. A context-count-only rule reaches 0.396 with high variance. Together, these controls support the relevance of the current-policy query while leaving the finer membership-versus-count attribution unresolved.

Random and TD-error controls are backbone dependent. In FPP DDPG, count-preserving Random-Support nearly matches SABER (0.604 versus 0.609 AUC), while a fixed random count falls to 0.540. In FPP SAC, Random-Support remains at 0.464 and fixed-count random weighting reaches 0.505, below SABER's 0.565 AUC despite saturated final success for the latter. PER-style loss weighting reaches 0.568 in FPP DDPG and 0.715 in DrawerOpen DDPG, showing that generic critic reweighting can itself be strong. SABER should therefore be presented as a support-conditioned allocation rule, not as proof that support alignment is the unique source of every gain.

Compute scaling answers a different question. FPP DDPG and TD3 high-replay-ratio 4x runs match SABER's AUC (0.618 and 0.449), while using four updates per environment step. SAC high-RR 4x reaches 0.739 AUC, above SABER's 0.565, and a 10-critic REDQ-HER target-Q-bound reference reaches 0.838. Their recorded mean training times are 16.1, 8.8, 47.0, and 89.5 hours respectively, versus 6.7, 6.7, and 9.6 hours for the corresponding SABER backbones. These are not hardware-normalized cost comparisons, but they rule out a claim that SABER dominates update or architecture scaling in environment-step AUC.

Table 2. Selected control closure. Values are mean success AUC on FPP unless the task is named explicitly.

Question	Compared conditions	Result	Supported interpretation
Does TD-error goal ranking reproduce the DDPG gain?	HGR / HER / SABER	0.338 / 0.539 / 0.609	Not in the matched FPP HGR experiment

Question	Compared conditions	Result	Supported interpretation
Is current-policy querying necessary?	Replay-query / shuffled-policy / SABER	0.368 / 0.347 / 0.442	Policy-query semantics matter in FPP TD3
Is exact membership assignment isolated?	Eligible count-matched / SABER	0.391 / 0.442; CI for delta includes 0	Directional, not causally confirmed
Is arbitrary fixed-count weighting sufficient?	DDPG fixed-count / HER / SABER	0.540 / 0.539 / 0.609	No in FPP DDPG
Can count-preserving randomization be strong?	DDPG Random-Support / SABER	0.604 / 0.609	Yes; exact alignment is not isolated in this row
Can more compute outperform SABER?	SAC high-RR / REDQ-HER / SABER	0.739 / 0.838 / 0.565	Yes, under larger update or critic budgets

H3: Support Is Error-Enriched, Not a Target-Quality Oracle

FPP diagnostics instrument 500 minibatches and 64k logged sample rows per seed for each backbone. Support-positive membership is weakly correlated with absolute TD error (Pearson 0.116--0.143), so it is not a surrogate for simply ranking the largest residuals. It is nevertheless enriched in the top-error region: top-10% support overlap is 0.516, 0.360, and 0.342 against support base rates 0.367, 0.219, and 0.197 for DDPG, TD3, and SAC. TD3 and SAC therefore show 1.67x and 1.83x top-error enrichment.

The same-minibatch update reduces support-positive residuals more than support-negative residuals. For TD3 the raw reduction is 0.021 versus 0.006 and remains larger after normalizing by the applied weight (0.013 versus 0.008). SAC shows 0.022 versus 0.006 raw and 0.013 versus 0.007 per-weight correction. DDPG also shows larger raw correction, but its per-weight values are close, 0.013 versus 0.011. These are local optimization responses on the weighted training minibatch, not held-out generalization measurements.

The target-variance result rejects a tempting stronger story. Support-positive target variance is higher, not lower, in all three FPP backbones. SABER should therefore not be justified as selecting intrinsically low-noise Bellman targets. The evidence supports error allocation and immediate correction, not a target-quality oracle.

DrawerOpen diagnostics provide a task-level view. Among the nine diagnostic seeds that reach Hit90, support-positive pre-update absolute TD error is 1.17x the negative-group value and is larger in all 9/9 seeds. Same-minibatch residual reduction is 2.09x and is also larger in 9/9 seeds. Allocation is phase dependent: the support-positive rate is 73.6% in approach, 70.5% in contact, 64.1% in opening, and 30.5% in the relabeled-goal-defined open phase. The one failed seed eventually has support near one for almost every sample, showing that support saturation is not sufficient for exploration or success.

Figure 3 (planned). A compact diagnostic figure should show, for FPP DDPG/TD3/SAC, support base rate versus top-10% overlap and positive/negative residual correction; a final DrawerOpen panel should show phase-conditioned support rate and mark the failed seed's late support saturation. The caption must state that residual correction is measured on the same minibatch.

Scope and Negative Results

Across the 13 completed task-backbone cells in the repository, SABER has higher mean AUC in 11 and lower mean AUC in two. This is a descriptive landscape, not a pooled statistical test: protocols and context adapters differ, and several cells have high variance. The two negative cells are FetchPush SAC (0.765 versus 0.783) and FetchSlide DDPG (0.301 versus 0.324). FetchPush TD3 also shows PER-style methods outperforming SABER despite a positive SABER-

HER delta. DrawerClose controls frequently match or exceed SABER in final success or AUC. An unmatched FPP DDPG SABER-only extension over seeds 6--10 also contains one collapsed seed and has AUC 0.477 +/- 0.207, substantially more variable than the primary seeds-1--5 cohort; without matched HER extension seeds, it is robustness evidence rather than an effect estimate. These results define the empirical boundary: support-conditioned critic allocation can help, but the gain depends on task dynamics, backbone, and the context predicate.

Runtime Cost

A controlled seed-1 FPP TD3 profile over the 20k--100k window measures 124.0 training steps/s for HER and 41.4 for SABER. SABER's training-loop time is 199% higher, and support lookup and weighting account for 62.3% of its measured loop. The profile is hardware specific and must not be extrapolated linearly to one million steps. Full-run throughput shows the same qualitative cost. SABER is an environment-step sample-efficiency method with a currently expensive support implementation, not a wall-clock efficiency result.

7. Limitations

First, the local context and phase definitions are task designed. The method is algorithmically reusable across backbones, but the present implementation is not task agnostic. Second, action bins provide coarse membership rather than a calibrated density; adding an eight-bin diversity threshold hurts FPP TD3, showing that a seemingly stronger support definition is not automatically better. Third, the scan uses at most 512 candidates in the main experiments. The DrawerOpen diagnostic cohort unexpectedly used the Python fallback rather than the requested Numba scanner, and its audit establishes implementation agreement only under the same candidate cap, not agreement with a full scan.

Fourth, mechanism attribution remains incomplete. The rigorous context-eligible count-matched control has a directionally lower AUC but an interval that includes zero, and random or TD-error reweighting is strong in several tasks. Fifth, the diagnostics are observational and same-minibatch; they do not establish that immediate residual correction causes downstream success. Sixth, the experiments cover simulated sparse manipulation rather than visual input, real robots, or broad continuous-control suites. Finally, support lookup materially reduces throughput, while high replay ratios and REDQ can match or exceed SABER's environment-step AUC with larger compute.

8. Conclusion

HER decides which hindsight targets are constructed, but standard implementations still allocate equal critic pressure after relabeling. SABER exposes this post-relabeling allocation decision and instantiates it with a local current-policy action-support predicate. Under matched HER sampling and a fixed one-update budget, SABER improves the primary FPP and DrawerOpen comparisons, including a paired 10-seed TD3 confirmation. The broader evidence also limits the conclusion: random and count-matched controls can be competitive, larger compute can be stronger, the support signal is not a low-variance oracle, and two task-backbone cells are negative. The resulting contribution is a bounded one: critic-loss allocation after hindsight relabeling is an empirically important design axis, and support-conditioned weighting is a useful but task-dependent instance of it.

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